



GLOBAL LAYOFF ANALYSIS

A Data Analysis Project on Global Layoffs Using Python

Abstract

This project focuses on analysing global layoffs trends using Python-based data analysis techniques. The dataset was cleaned and processes to ensure accuracy, followed by exploratory data analysis to identify patterns across companies, industries, and countries.

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Global Layoff Analysis

Overview

The Global Layoff Analysis Project focuses on analysing layoff trends across different companies, industries, and countries using Python-based data analysis.

The project is structured in a modular and organised way to ensure scalability, readability, and reproducibility. It includes data processing, cleaning, analysis, and visualization using standard data analysis libraries.

Business Problems

- What are the overall layoff trends globally?
- Which companies are most affected by layoffs?
- Which industries experienced the highest layoffs?
- How do layoffs vary across countries?
- What is the trend of layoffs over time?

Dataset Information

Source: Global Layoffs Dataset (Kaggle, CSV format)

The dataset is suitable for analysing global layoff trends, enabling insights into company-level, industry-level, and country-level workforce changes over time.

Time Period: 2020 - 2026

Total Records:

- Before Cleaning: 4,305 rows
- After Cleaning: 3602 rows

Total Columns: 11 features

Project Structure

The project is organised into multiple folders to ensure a clear separation of data, analysis, and supporting files.

- 1_Data/ - Store raw and cleaned datasets
- 2_Notebooks/ - Main analysis work (Jupyter Notebook)
- 3_Images - All generated visualizations
- 4_src/ - Source code for reusable functions
- 5_README.md - Project summary for GitHub
- 6_Requirements.txt - Project dependencies
- 7_.ignore - Files and folders excluded from version control
- 8_Global_Layoff_Analysis.pdf - Detailed project documentation

Tools & Technologies used

- Python - Core programming language used for data analysis
- Pandas - Used for data manipulation, cleaning, and analysis
- Matplotlib - Used for creating visualizations and charts
- Jupyter Notebook - Used as the development environment for analysis and documentation

Data Cleaning and Analysis

Data Loading and Initial Exploration

The analysis began by importing the required libraries:

- **Pandas** for Data Manipulation
- **Matplotlib** for Visualization

The dataset was loaded using `pd. read_csv ()` into a DataFrame for further processing.

Initial exploration of the dataset was performed using:

- `. head ()` - to preview the data
- `. Shape` - to understand data size
- `. info ()` - to inspect data types and missing values
- `. describe ()` - to generate statistical summary

Data Cleaning

To ensure data quality and reliability, several preprocessing steps were performed:

- **Missing Value Analysis**
Checked for null values using `.isnull().sum()`
Focused on key columns:
 - `total_laid_off`
 - `percentage_laid_off`
- **Duplicate Records Handling**
 - Identify duplicate rows using `.duplicated().sum()`
- **Standardization of Text Data**
Cleaned string columns using `.str.strip()`
 - `Company`
 - `Location`
 - `Country`
- **Handling Inconsistent Values**
Standardized country names:
 - Replaced "UAE" with "United Arab Emirates"
Verified using `.unique()`
- **Data Filtering**
Checked records where:
 - `total_laid_off` is null
 - `percentage_laid_off` is null

Remove Irrelevant records:

- Dropped rows where both values were missing, as they do not contribute to meaningful analysis

Reset index after cleaning:

- `df.reset_index(drop=True)`
- **Data Export**
The cleaned dataset was saved for further analysis:
 - Stored as a CSV file in the `clean_data` folder using `.to_csv()`

Exploratory Data Analysis (EDA)

Company – Level Analysis

To identify companies most affected by layoffs:

- Sorted the dataset based on total layoffs in descending order
- Extracted the top 10 records with highest layoffs

Additionally:

- Performed aggregation using:
 - Groupby ('company')
 - Calculated total layoffs using. sum ()
- Sorted results in descending order
- Selected top 10 companies with highest layoffs

This helps identify major contributors to global layoffs

Industry -Level Analysis

To understand layoffs across different industries:

- Used .value_counts () on the industry column
 - Identify the top 10 industries by frequency
- Performed aggregation:
 - Groupby ('industry')
 - Calculated total layoffs using. sum ()
- Sorted values in descending order
- Extracted top 10 industries with highest layoffs

This helps analyse which sectors were most impacted.

Country-Level Analysis

To analyse layoffs across different countries:

- Grouped the dataset by country using:
 - Groupby('country')
- Calculated total layoffs using:
 - . sum () on total_laid_off
- Sorted the result in descending order (ascending=False)
- Extracted the top 10 countries with the highest layoffs

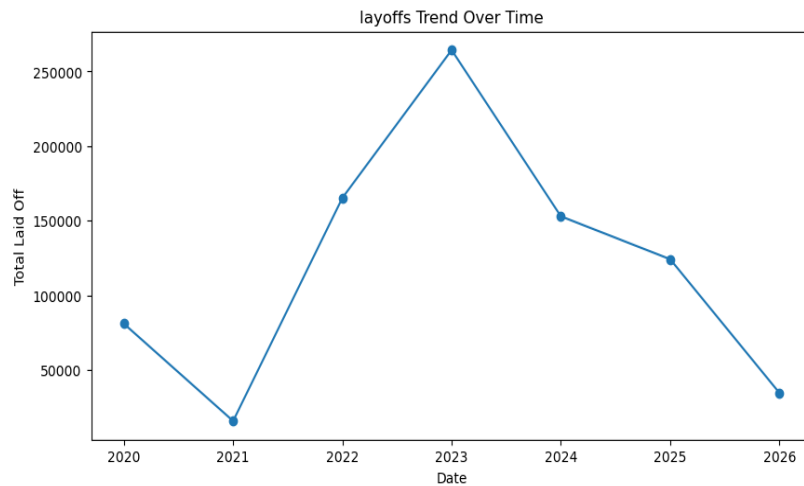
This analysis highlights the countries most impacted by layoffs and helps identify global patterns in workforce reductions.

Visualizations

Layoff Trend Over Time (Year-wise)

- Extracted year from date column using:
 - `df['year']=df['date'].dt.year`
- Grouped data by year and calculated total layoffs
- Created a line chart to visualize overall layoff trends

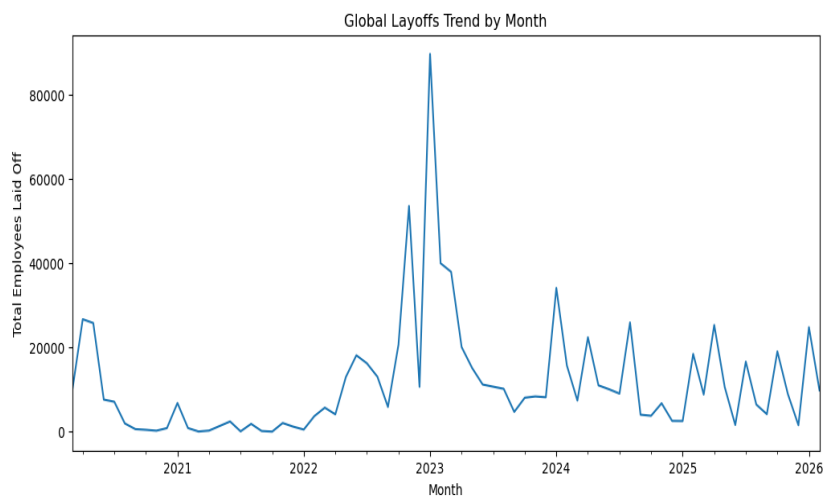
Helps identify long-term trends and peak layoff periods



Monthly Layoff Trend

- Extracted month using:
 - `df['month']=df['date'].dt.to_period('M')`
- Aggregated layoffs by month
- Visualized using a line chart

Shows short-term fluctuations and monthly patterns

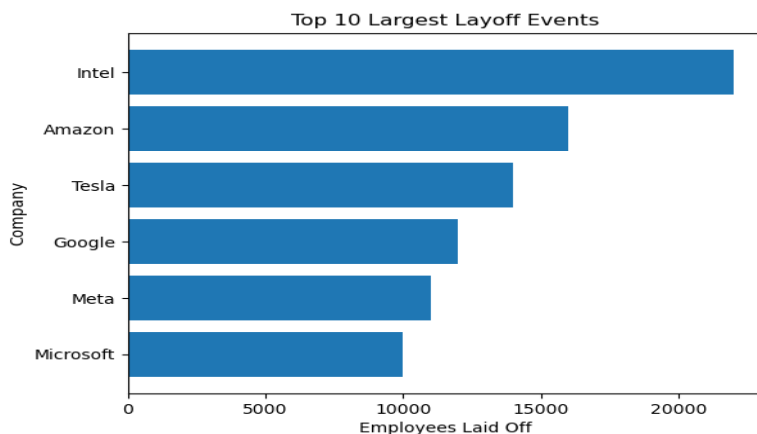


Company-Level Visualization

Top 10 Largest Layoff Events

- Sorted dataset by total_laid_off in descending order
- Selected top 10 records
- Created a horizontal bar chart

Highlights companies with the most significant layoff events

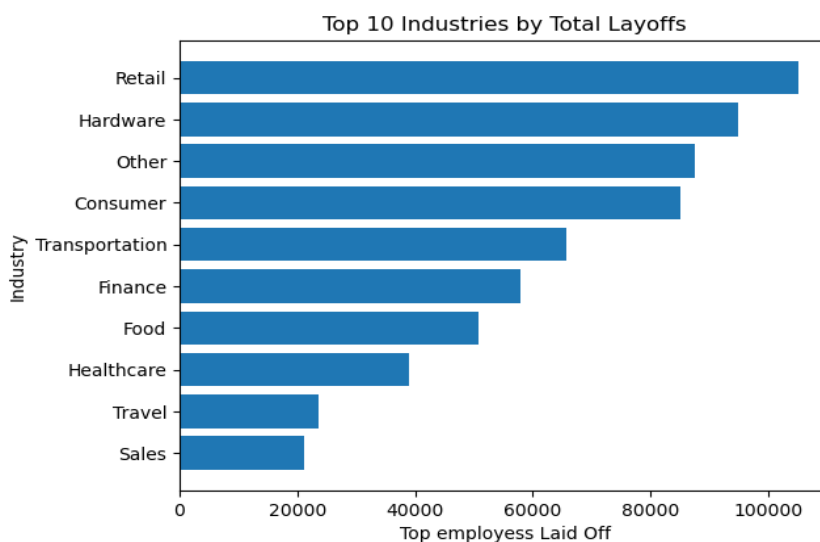


Industry-Level Visualization

Top 10 Industries by Layoffs

- Grouped data by industry and calculated total layoffs
- Selected top 10 industries
- Visualized using a horizontal bar chart

Identifies most affected sectors

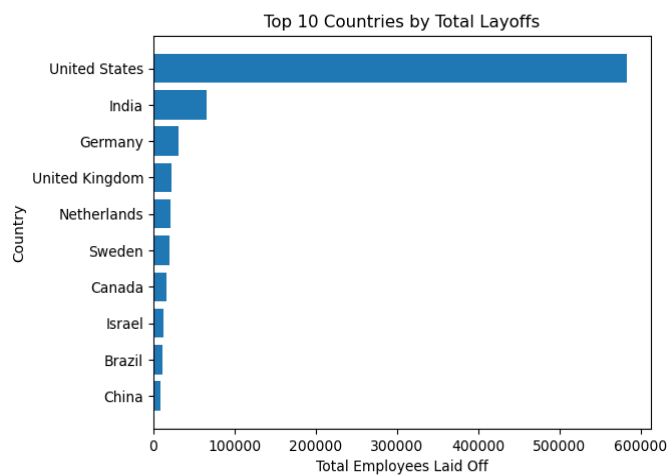


Country-level Visualization

Top 10 Countries by Layoffs

- Grouped data by country and calculated total layoffs
- Extracted top 10 countries
- Created a horizontal bar chart

Shows geographical distribution of layoffs

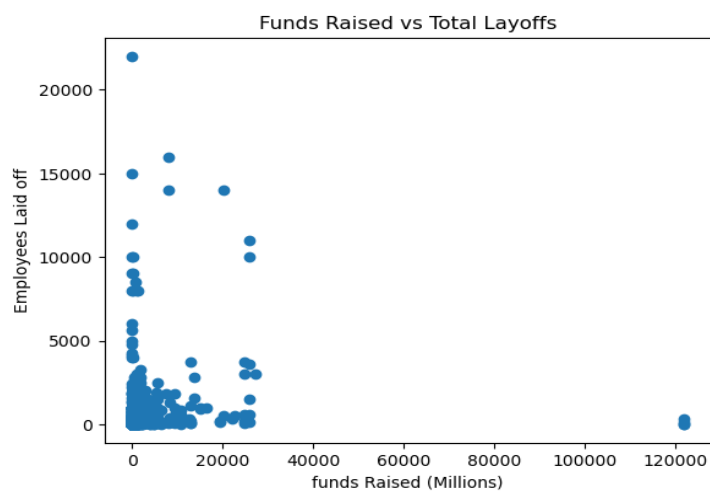


Additional Analysis

Funds Raised vs Layoffs

- Analysed relationship between
 - funds_raised
 - total_laid_off
- Created a scatter plot

Helps understand whether higher funding correlates with layoffs



Key Insights

1. Layoffs Trends Over Time

- Layoffs significantly increased over the years, with a major peak in:
 - 2023 (~264K layoffs)
- A sharp rise is also observed in:
 - 2022 and 2024
- Early years like 2020 and 2021 had comparatively lower layoffs

Indicates a post-pandemic economic correction and restructuring phase

2. Monthly Trend Analysis

- Layoffs show high fluctuations month-to-month
- A major spike is observed around early 2023
- No consistent seasonal pattern, but sudden spikes suggest:
 - Company restructuring waves
 - Economic uncertainty

3. Top Companies with Highest Layoffs

- Intel (~22,00 layoffs) recorded the highest single layoff event
- Followed by:
 - Amazon
 - Tesla
 - Google
 - Meta
 - Microsoft

Major layoffs are concentrated in large tech companies

4. Industry-wise Impact

- Retail (~105K) layoff is the most affected industry
- Followed by:
 - Hardware
 - Consumer
 - Transportation

Indicates strong impact on:

- Consumer-facing industries
- Supply chain & hardware sectors

5. Country-wise Impact

- United States (~583K layoffs) dominates significantly
- Followed by
 - India
 - Germany
 - United Kingdom

Shows layoffs are heavily concentrated in tech-driven economies

6. Funding vs Layoffs Relationship

- Correlation between funds raised and layoffs = ~ 0.11

This indicates:

- Very weak relationship
- Even well-funded companies conducted layoffs

Suggests layoffs are driven by:

- Cost optimisation
- Strategic restructuring
- Not just lack of funding

Recommendations

- Companies should closely monitor layoff trends over time to better understand how economic conditions impact workforce decisions.
- Industries such as Retail and Hardware may require more stable operational strategies, as they show higher layoffs concentrations.
- Since layoffs are concentrated in specific regions, organizations can analyse geographic risks before expanding or restructuring.

Conclusion

- The analysis reveals clear patterns in layoffs across Countries, Industries, and time periods.
- The United States and major tech-driven sectors show the highest impact, indicating regional and industry concentration.
- A significant rise in layoffs during 2022-2023 highlights the effect of broader economic changes.
- The weak correlation between funding and layoffs suggests that financial strength alone does not guarantee workforce stability.
- Overall, this project demonstrates how data analysis helps in understanding trends and supporting better decision-making, rather than predicting outcomes.